

Linearly independent solutions

Linear dependent functions

A set of functions $f_1(x), f_2(x), \dots, f_n(x)$ is linearly dependent on a given interval if there exist constants $c_1, c_2, \dots, c_n$ not all zero such that

$$c_1 f_1(x) + c_2 f_2(x) + \dots c_n f_n(x) = 0$$

Functions are linearly independent if there are not linearly dependent.

If $f_1(x), f_2(x), \dots, f_n(x)$ have $n-1$ derivatives. The determinant

$$W = \begin{vmatrix} f_1 & f_2 & \dots & f_n \\ f_1' & f_2' & \dots & f_n' \\ \vdots & \vdots & \dots & \vdots \\ f_1^{n-1} & f_2^{n-1} & \dots & f_n^{n-1} \end{vmatrix}$$

is called the **Wronskian** of the functions.

Value of the Wronskian can help us to determine if the functions are dependent or independent.

If $y_1(x), y_2(x)$ are two solutions of a second order differential equation and $W(y_1, y_2) \neq 0$ then y_1 and y_2 are called fundamental set of solutions and $y = c_1 y_1 + c_2 y_2$ is the general solution.

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \neq 0$$

Example 1

Find the general solution of the differential equation

$$(D^2 - D - 2)y = 0$$

$$y = e^{mx}$$

$$(m + 1)(m - 2) = 0$$

$$m = -1, m = 2$$

$$y = Ae^{-x} + Be^{2x}$$

Check these solutions are linearly independent.

$$\text{Wronskian} = W = \begin{vmatrix} e^{-x} & e^{2x} \\ -e^{-x} & 2e^{2x} \end{vmatrix} = 3e^x \neq 0$$

(Linearly independent)

General solution $y = Ae^{-x} + Be^{2x}$

Example 2

$y_1 = e^{-2t}$, $y_2 = e^{-3t}$ are the roots of the equation

$$(D^2 + 5D + 6)y = 0 .$$

$$W = \begin{vmatrix} e^{-2t} & e^{-3t} \\ -2e^{-2t} & -3e^{-3t} \end{vmatrix} = -e^{-5t} \neq 0$$

$\therefore y_1, y_2$ give the solutions of the above differential equation.

Example 3

Obtain the solutions of the equation $(D^2 - D - 2)y = 0$

$$y = e^{mx}$$

$$(m + 1)(m - 2) = 0$$

$$m = -1, m = 2$$

$$y = Ae^{-x} + Be^{2x}$$

Let us check whether they are Linearly independent.

$$\text{Wronskian} = W = \begin{vmatrix} e^{-x} & e^{2x} \\ -e^{-x} & 2e^{2x} \end{vmatrix} = 3e^x \neq 0$$

$\therefore$ General solution $y = Ae^{-x} + Be^{2x}$.